

Ethan Yang

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Education

University of Toronto

Expected Graduation: May 2030

Bachelor of Applied Science in Computer Engineering

Toronto, Ontario

- GPA: 3.94
- Courses: Software Engineering Practice, Applied Machine Learning, Operating Systems, Computer Architecture, Model-Based Programming, Algorithms and Data Structures, Intro. to Software Engineering

Work Experience

Sun Life Financial

May 2026 – Present

API Development Intern

Toronto, Ontario

- Developed and maintained **RESTful APIs** on Sun Life's **Enterprise API Platform** by adhering to **OpenAPI** specifications and internal governance standards, supporting digital transformation across business units

MannLab

January 2026 – Present

Undergraduate Researcher

Toronto, Ontario

- Co-authored **2 research papers** on **EEG biofeedback** and **consumer BCI safety** by conducting **wearable computing** and **XR** research under Professor Steve Mann, submitted to **IEEE ISMAR** and **NeurIPS**
- Promoted **cold-exposure therapy** as a neurochemical substitute for cocaine by building *Cocaine Buzzkill*, an **XR** game on **Meta Quest 3** with real-time **Muse S EEG** biofeedback modulating gameplay parameters
- Identified unsafe **TTA** adaptation patterns by auditing **8 methods** across **9,144 EEG windows** and implementing a **3-pass activation patching** diagnostic in **PyTorch** to detect artifact-mediated failure circuits pre/post adaptation

Robotics for Space Exploration

September 2025 – April 2026

Software Developer

Toronto, Ontario

- Reduced the risk of unsafe arm commands during competition by developing and **unit testing ROS2** communication safety nodes for a **robotic arm** in **Python** with fault-tolerant inter-process communication
- Contributed to the team achieving **93.96%** on the **University Rover Challenge SAR (20+ points)** above the qualification cutoff) by building a **PyQt** rover control **GUI** integrated with **ROS2**

Projects

Finance Digitization | [Source Code](#)

Next.js | **TypeScript** | **AWS** | **PostgreSQL**

- Digitized cheque requisitions for **UofT EngSoc** by building a full-stack web app with **OCR**-based receipt parsing, automated form filling, and a **dual e-signature** approval workflow

Sinatra | [Source Code](#)

React | **TypeScript** | **FastAPI** | **Python**

- Built a browser-based **DAW** converting vocals to **MIDI** instruments using **ML** transcription, with **AI** chord generation and **Three.js 3D** video export; demoed to **250 people** with a waitlist of **40**

HATSEYE — DeltaHacks 12 Finalist | [Source Code](#) | [Demo](#)

Python | **Flask** | **OpenCV** | **Arduino**

- Placed **top 6 of 144 teams** by building a voice-activated **AI** vision assistant for visually impaired users using **Gemini Vision** with real-time **haptic** obstacle feedback via **ultrasonic sensors** and **Arduino**

Lock Block — HackHive 2026 Winner | [Source Code](#) | [Demo](#)

Python | **Flask** | **OpenCV** | **Arduino**

- Secured homes against unauthorized entry by building a smart lock system using **OpenCV** facial recognition that auto-locks on unknown faces and enables remote unlock via **blockchain** wallet authentication

Leadership

UofTHacks

March 2026 – Present

Event Logistics Executive

Toronto, Ontario

- Organizing hacker mini-events and external sponsor/club collaborations for **UofTHacks 14** by planning event flows, coordinating with external guests, and managing room allocations and scheduling across the hackathon

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, SQL, C, C++, HTML5, CSS

Libraries/Frameworks: React, Next.js, NodeJS, FastAPI, Flask, PyTorch, OpenCV, PyQt

Developer Tools: Git, AWS, Docker, Postman, Jira, ROS2, Arduino